Solution

1 The problem

Question 1. Let $\mathbb{H}$ be a hyperfield with multivalued addition operator $\boxplus$. Let $x, y, z \in \mathbb{H}$ be elements. Suppose that $0 \in x \boxplus y \boxplus z$. Does it then follow that

$$0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus -xyz \boxplus -xyz \boxplus -xyz\,?$$

2 The result

The answer is *no*.

Theorem 1. *There is a hyperfield $\mathbb{H}$ with elements x, y, z such that*

$$0 \in x \boxplus y \boxplus z \qquad \textit{but} \qquad 0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

One may take $\mathbb{H}$ infinite, with multiplicative group $\mathbb{H}^\times = \langle t \rangle \cong \mathbb{Z}$ infinite cyclic, and $x = t$, $y = t^4$, $z = 1$.

Conventions. "Hyperfield" is meant in the standard sense of Krasner [4, 5]; see also [6, 1]. Thus a hyperfield is a set $\mathbb{H}$ with a distinguished element 0 and the following structure:

- a commutative, associative hyperoperation $\boxplus$ with nonempty values, for which 0 is neutral;
- unique negatives: for each a there is exactly one b, denoted $-a$, with $0 \in a \boxplus b$;
- reversibility: $c \in a \boxplus b \Rightarrow a \in c \boxplus (-b)$;
- a multiplication making $\mathbb{H} \setminus \{0\}$ an abelian group, with $0 \cdot h = h \cdot 0 = 0$ and $w(a \boxplus b) = wa \boxplus wb$.

Sums of sets are taken elementwise, $A \boxplus B = \bigcup_{a \in A,\, b \in B} a \boxplus b$, and $a \boxplus B := \{a\} \boxplus B$. By associativity, iterated sums do not depend on the bracketing (nor, by commutativity, on the order of the summands).

3 Overview of the proof

When $\mathbb{H}$ is a quotient $F/K^\times$ of a field, the answer is yes, because of the identity $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$ (Remark 8). A counterexample must therefore come from a hyperfield that is not of this kind, and we use hyperfields built from projective planes.

Given an abelian group G acting regularly on the points of a projective plane by collineations, with at least five points on each line, set $\mathbb{H} = G \cup \{0\}$, $a \boxplus a = \{0, a\}$ and $a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\}$ for $a \neq b$, where $\mathrm{L}(a, b)$ is the line through a and b. This is a hyperfield in which every element is its own negative (Proposition 3); a construction of this kind appears in the work of Connes and Consani [2]. In such an $\mathbb{H}$, with $z = 1$, the hypothesis $0 \in x \boxplus y \boxplus 1$ holds as soon as $1, x, y$ are

distinct and collinear, while the six-term sum equals $(\mathcal{M} \cup \{0\}) \boxplus 1$, where $\mathcal{M}$ is the line through x^3, y^3, xy (if these are distinct). This sum avoids 0 whenever $1 \notin \mathcal{M}$ (Lemma 4). So we need a plane in which $1, x, y$ are collinear but 1 is not on the line through x^3, y^3, xy; by the remark above, no quotient of a field can serve here.

Such a plane is obtained on the point set $\mathbb{Z}$ (written additively, so that n stands for t^n) from an infinite perfect difference set $D \subset \mathbb{Z}$, whose translates $g + D$ serve as lines (Lemma 6). A greedy construction (Lemma 7), in the spirit of M. Hall's infinite cyclic planes [3], produces such a D containing the seed $\{0, 1, 4, 103, 105, 112\}$ and not containing 100. Then $0, 1, 4$ lie on the line D, the points $3, 12, 5$ (corresponding to x^3, y^3, xy) lie on the line $-100 + D$, and 0 does not.

All arguments below are given in full; the only external input is the definition of a hyperfield.

4 Hyperfields from projective planes with a regular abelian group

Let G be an abelian group, written multiplicatively. Consider an incidence structure with point set G and a family of subsets of G, called *lines*, satisfying:
(P1) any two distinct points lie on exactly one line;
(P2) any two distinct lines meet;
(P3) every line has at least 5 points;
(P4) for every $w \in G$, multiplication by w maps lines to lines.
By (P1) two distinct lines have at most one common point, and by (P2) they have at least one; so two distinct lines meet in exactly one point. For distinct points a, b we write $\mathrm{L}(a, b)$ for the line through a and b.

Definition 2. Put $\mathbb{H} = G \cup \{0\}$. Let 0 be neutral: $0 \boxplus a = a \boxplus 0 = \{a\}$ for all $a \in \mathbb{H}$ (so $0 \boxplus 0 = \{0\}$). For $a, b \in G$ set

$$a \boxplus a = \{0, a\}, \qquad a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\} \quad (a \neq b).$$

Multiplication is the group law on G, extended by $0 \cdot h = h \cdot 0 = 0$.

Proposition 3. *$\mathbb{H}$ is a hyperfield, and $-a = a$ for every $a \in \mathbb{H}$.*

Proof. Commutativity and nonempty values. Commutativity is clear. For $a \neq b$ in G, the set $a \boxplus b$ has at least 3 points by (P3), so every value is nonempty.

Negatives. Let $a \neq 0$. For $b \in G$ with $b \neq a$ we have $a \boxplus b \subseteq G$, so $0 \notin a \boxplus b$. Also $0 \boxplus a = \{a\}$ does not contain 0, while $0 \in a \boxplus a$. Hence a is the unique negative of a. The element 0 is its own unique negative.

Reversibility. Since $-b = b$, we must show $c \in a \boxplus b \Rightarrow a \in c \boxplus b$.

- If $a = 0$, then $c = b$ and $0 \in b \boxplus b$; this includes the case $b = 0$.
- If $b = 0$, then $c = a$ and $a \in a \boxplus 0$.
- If $a = b \neq 0$, then $c \in \{0, a\}$. If $c = 0$, then $a \in 0 \boxplus a$. If $c = a$, then $a \in a \boxplus a$.
- If $a \neq b$, both nonzero, then $c \in \mathrm{L}(a, b) \setminus \{a, b\}$. Since $c \neq b$, $\mathrm{L}(c, b) = \mathrm{L}(a, b)$, so $c \boxplus b = \mathrm{L}(a, b) \setminus \{c, b\}$. This set contains a, because $a \neq c$ and $a \neq b$.

Distributivity. The case $w = 0$ is trivial, so let $w \in G$. Multiplication by w is a bijection of G that maps lines to lines by (P4). Hence $w\,\mathrm{L}(a, b)$ is a line through wa and wb, i.e. $w\,\mathrm{L}(a, b) = \mathrm{L}(wa, wb)$, and so $w(\mathrm{L}(a, b) \setminus \{a, b\}) = \mathrm{L}(wa, wb) \setminus \{wa, wb\}$. Also $w\{0, a\} = \{0, wa\} = wa \boxplus wa$, and the cases involving 0 are trivial.

Associativity. If one of a, b, c is 0, both $(a \boxplus b) \boxplus c$ and $a \boxplus (b \boxplus c)$ equal the sum of the other two, because 0 is neutral.

Now let a, b, c be nonzero and put $S(a,b,c) := (a \boxplus b) \boxplus c$. We show that S depends only on the multiset $\{a,b,c\}$. Then, using commutativity,

$$a \boxplus (b \boxplus c) = (b \boxplus c) \boxplus a = S(b,c,a) = S(a,b,c) = (a \boxplus b) \boxplus c.$$

Case 1. $a = b = c$. Then $S = (0 \boxplus a) \cup (a \boxplus a) = \{0, a\}$.

Case 2. The multiset is $\{a, a, c\}$ with $a \neq c$; let $L = \mathrm{L}(a,c)$. Up to commutativity there are two orderings.

- $(a \boxplus a) \boxplus c = (0 \boxplus c) \cup (a \boxplus c) = \{c\} \cup (L \setminus \{a,c\}) = L \setminus \{a\}$.
- $(a \boxplus c) \boxplus a = \bigcup_{p \in L \setminus \{a,c\}} (p \boxplus a) = \bigcup_{p \in L \setminus \{a,c\}} (L \setminus \{p,a\})$. The point a lies in none of these sets. A point $q \in L \setminus \{a\}$ is missing from $L \setminus \{p,a\}$ only when $p = q$, and there are at least 3 choices of p. So the union is $L \setminus \{a\}$.

Case 3. a, b, c are distinct and lie on a common line L. Here $a \boxplus b = L \setminus \{a,b\}$, which contains c. Then

$$S = (c \boxplus c) \cup \bigcup_{p \in L \setminus \{a,b,c\}} (L \setminus \{p,c\}) = \{0,c\} \cup (L \setminus \{c\}) = L \cup \{0\},$$

where we used that $L \setminus \{a,b,c\}$ has at least 2 points, by (P3). The result is symmetric in a, b, c.

Case 4. a, b, c are not collinear. We claim

$$S = G \setminus \big(\mathrm{L}(a,b) \cup \mathrm{L}(b,c) \cup \mathrm{L}(a,c)\big),$$

which is symmetric in a, b, c. By definition,

$$S = \bigcup_{p \in \mathrm{L}(a,b) \setminus \{a,b\}} \big(\mathrm{L}(p,c) \setminus \{p,c\}\big);$$

here $p \neq c$ because $c \notin \mathrm{L}(a,b)$. Also $\mathrm{L}(p,c) \neq \mathrm{L}(a,b)$, so $\mathrm{L}(p,c) \cap \mathrm{L}(a,b) = \{p\}$.

- $0 \notin S$, and $c \notin S$.
- *Points of* $\mathrm{L}(a,b)$ *are not in* S (including a and b): such a point lies in $\mathrm{L}(p,c)$ only if it equals p, and p is excluded.
- *Points* $q \in \mathrm{L}(a,c) \setminus \{a,c\}$ *are not in* S. If $q \in \mathrm{L}(p,c) \setminus \{p,c\}$, then $\mathrm{L}(p,c) = \mathrm{L}(q,c) = \mathrm{L}(a,c)$. So $p \in \mathrm{L}(a,c) \cap \mathrm{L}(a,b) = \{a\}$, since these lines are distinct; this contradicts $p \neq a$.
- *Points* $q \in \mathrm{L}(b,c) \setminus \{b,c\}$ *are not in* S, by the same argument with b in place of a.
- *Every other point* q *of* G *is in* S. Such a q lies on none of the three lines. The lines $\mathrm{L}(q,c)$ and $\mathrm{L}(a,b)$ are distinct because $c \notin \mathrm{L}(a,b)$, so by (P2) they meet in a point p. Then $p \neq a$, since otherwise $q \in \mathrm{L}(a,c)$; likewise $p \neq b$. Also $p \neq q$, since $q \notin \mathrm{L}(a,b)$, and $p \neq c$. So $q \in \mathrm{L}(p,c) \setminus \{p,c\}$ with $p \in \mathrm{L}(a,b) \setminus \{a,b\}$, hence $q \in S$.

These cases cover every multiset of three nonzero elements, so $\boxplus$ is associative. □

5 The six-term sum in such a hyperfield

Lemma 4. *Let* $\mathbb{H}$ *be as in Section 4, and let* $x, y \in G$ *satisfy:*

(i) *the points* $1, x, y$ *are distinct and collinear;*

(ii) *the points x^3, y^3, xy are distinct and lie on a line $\mathcal{M}$ with $1 \notin \mathcal{M}$.*
Then with $z = 1$ we have $0 \in x \boxplus y \boxplus z$, but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$.

Proof. The hypothesis holds. Since $1, x, y$ are distinct and collinear, $1 \in \mathrm{L}(x, y) \setminus \{x, y\} = x \boxplus y$. Hence $(x \boxplus y) \boxplus 1 \supseteq 1 \boxplus 1 \ni 0$.

The conclusion fails. Every element is its own negative (Proposition 3), so $-xyz = xy$; also $z^3 = 1$. The six-term sum is therefore $T \boxplus 1$, where

$$T := x^3 \boxplus y^3 \boxplus xy \boxplus xy \boxplus xy.$$

First, $xy \boxplus xy \boxplus xy = \{0, xy\} \boxplus xy = \{xy\} \cup \{0, xy\} = \{0, xy\}$. So

$$T = (x^3 \boxplus y^3) \boxplus \{0, xy\} = (x^3 \boxplus y^3) \cup \big((x^3 \boxplus y^3) \boxplus xy\big).$$

The first part is $\mathcal{M} \setminus \{x^3, y^3\}$. The second part is $\mathcal{M} \cup \{0\}$, by Case 3 of the proof of Proposition 3, since x^3, y^3, xy are distinct and lie on $\mathcal{M}$. Hence $T = \mathcal{M} \cup \{0\}$.

Finally, $T \boxplus 1 = \bigcup_{u \in T} u \boxplus 1$. By uniqueness of negatives, $0 \in u \boxplus 1$ if and only if $u = -1 = 1$. Since $1 \notin \mathcal{M}$ and $1 \neq 0$, we have $1 \notin T$, so $0 \notin T \boxplus 1$. □

6 A plane containing the required configuration

From now on $G = \mathbb{Z}$, written additively: the integer n stands for the group element t^n, so the identity element 1 of G corresponds to $0 \in \mathbb{Z}$, and multiplication in G corresponds to addition in $\mathbb{Z}$.

Definition 5. A set $D \subset \mathbb{Z}$ is a *perfect difference set* if every nonzero integer has exactly one representation $d - d'$ with $d, d' \in D$. A set $D \subset \mathbb{Z}$ is *Sidon* if every nonzero integer has at most one such representation.

Lemma 6. *If $D \subset \mathbb{Z}$ is an infinite perfect difference set, then the lines $\{g + D : g \in \mathbb{Z}\}$ on the point set $\mathbb{Z}$ satisfy* (P1)–(P4).

Proof. (P1): Distinct points p, q lie on $g + D$ if and only if $p - g = d_1$ and $q - g = d_2$ for some $d_1, d_2 \in D$. Then $p - q = d_1 - d_2$. This representation of the nonzero integer $p - q$ exists and is unique, so $g = p - d_1$ exists and is unique.

(P2): Distinct lines are $g + D$ and $h + D$ with $g \neq h$. Let $h - g = d_1 - d_2$ be the representation. Then both lines contain the point $g + d_1 = h + d_2$.

(P3): Every line is infinite.

(P4): Translating $g + D$ by w gives $(g + w) + D$, which is again a line. □

Lemma 7. *There is an infinite perfect difference set $D \subset \mathbb{Z}$ with*

$$D \supseteq D_0 := \{0, 1, 4, 103, 105, 112\} \qquad \textit{and} \qquad 100 \notin D.$$

Proof. D_0 is Sidon. Its 15 positive differences are

$$\begin{gathered} 1 - 0 = 1, \quad 4 - 0 = 4, \quad 4 - 1 = 3, \quad 103 - 0 = 103, \quad 103 - 1 = 102, \quad 103 - 4 = 99, \\ 105 - 0 = 105, \quad 105 - 1 = 104, \quad 105 - 4 = 101, \quad 105 - 103 = 2, \\ 112 - 0 = 112, \quad 112 - 1 = 111, \quad 112 - 4 = 108, \quad 112 - 103 = 9, \quad 112 - 105 = 7, \end{gathered}$$

that is, $1, 2, 3, 4, 7, 9, 99, 101, 102, 103, 104, 105, 108, 111, 112$. These are pairwise distinct.

Inductive step. Let D_k be a finite Sidon set of nonnegative integers, and let

- m be the least positive integer that is not a difference of two elements of D_k;
- $R = \max D_k$;
- N be an integer with $N > 3R + m$.

Put $D_{k+1} = D_k \cup \{N, N+m\}$. These are new elements, since $N > R$. The new positive differences are $m = (N+m) - N$, the numbers $N - d$, and the numbers $N + m - d$, for $d \in D_k$. We check that all positive differences of D_{k+1} (counted with their representations) are distinct:

- Old positive differences are at most R. Every $N - d$ and every $N + m - d$ is at least $N - R > 2R + m$. So these exceed both the old differences and m.
- The values $N - d$ ($d \in D_k$) are pairwise distinct, and so are the values $N + m - d$.
- If $N - d = N + m - d'$, then $d' - d = m$, which would make m a difference of D_k; this is impossible.
- m itself is not an old difference, by its choice.

So D_{k+1} is a Sidon set of nonnegative integers, and m is now a difference.

Covering. Every difference of D_k remains a difference of D_{k+1}, and m becomes one. So the least positive non-difference strictly increases at each step. Hence every positive integer is eventually a difference, and by symmetry so is every negative integer.

The limit set. Let $D = \bigcup_k D_k$. It is infinite. Any two representations of the same nonzero integer as a difference of elements of D already lie in a common D_k, so they coincide. By covering, every nonzero integer has at least one representation. So D is a perfect difference set.

$100 \notin D$. We have $100 \notin D_0$, and every element added at a later step is at least $N > 3R \geq 3 \cdot 112 > 100$, as $R \geq \max D_0 = 112$ at every step. □

7 The counterexample

Proof of Theorem 1. Let D be as in Lemma 7. By Lemma 6, the lines $g + D$ on $\mathbb{Z}$ satisfy (P1)–(P4), so Proposition 3 gives a hyperfield $\mathbb{H} = \langle t \rangle \cup \{0\}$. Take $x = t$, $y = t^4$, $z = 1$; in additive notation $x \leftrightarrow 1$, $y \leftrightarrow 4$ and $1_{\mathbb{H}} \leftrightarrow 0$. We check the hypotheses of Lemma 4.

- The points $1_{\mathbb{H}}, x, y$ are $0, 1, 4$. They are distinct and all lie on the line D.
- The points x^3, y^3, xy are $3, 12, 5$. They are distinct and lie on the line $\mathcal{M} = -100 + D$, because $103, 112, 105 \in D$.
- $1_{\mathbb{H}}$ is not on $\mathcal{M}$: the point 0 lies on $\mathcal{M}$ if and only if $100 \in D$, which is false.

By Lemma 4, $0 \in x \boxplus y \boxplus z$ but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$. This proves Theorem 1, so the answer to Question 1 is no. □

Remark 8. The following observation is not used in the proof; it explains why a counterexample cannot be of the most familiar type. Let F be a commutative field and $K^\times$ a subgroup of $F^\times$, and consider Krasner's quotient hyperfield $F/K^\times$ [5], with classes $[X] = XK^\times$ and $[X] \boxplus [Y] = \{[X' + Y'] : X' \in [X], Y' \in [Y]\}$. Suppose $0 \in x \boxplus y \boxplus z$ there. Then some $w \in x \boxplus y$ satisfies $0 \in w \boxplus z$, so $w = -z$; hence there are representatives $X \in x$, $Y \in y$ with $X + Y \in -z$, and $Z := -(X + Y)$ is a representative of z with $X + Y + Z = 0$. The identity

$$X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$$

shows that $X^3 + Y^3 + Z^3 + (-XYZ) + (-XYZ) + (-XYZ) = 0$, and $X^3, Y^3, Z^3, -XYZ$ are representatives of $x^3, y^3, z^3, -xyz$; so 0 lies in the six-term sum. Heuristically, this is why the counterexample above uses a projective plane that is not of field origin. We do not prove here that the plane of Lemma 7 is non-Desarguesian, and nothing depends on it.

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